

Strategy Ranking

Honest Assessment of All 38 Strategies in alan_trader

Audit date: 2026-05-19 • Auditor: Claude Opus 4.7 (1M context)

Scope: 18 AI-driven + 20 rules-based

Top 10 Overall

Ranked by likelihood of making money in production — combining edge thesis strength, backtest realism, code quality, risk profile, and live-readiness.

#	Strategy	Type	Grade	Why it's here	Trades/yr
1	hmm_regime	AI	A	Classifies current vol regime (not predicting); matches defined-risk structure to it. Cleanest walk-forward in the codebase. Slippage modeled.	12–18
2	rates_spy_rotation_option	Rules	A	Four-regime rotation on real option chain data. Roll logic sound. Long-only options = capped downside.	8–12
3	vix_spike_fade	Rules	B+	VIX mean reversion into 200d MA is one of the most empirically reliable edges; bull call spread caps cost.	8–15
4	iron_condor_rules	Rules	B	Vanilla VRP harvest with three hard gates (IVR / VIX / ADX). 50% profit target. Production-grade simplicity.	8–15
5	fomc_event_straddle	Rules	B	Calendar-triggered (FOMC dates published, no lookahead). Realized > implied at events documented (Lucca, Savor).	~8
6	tail_risk_put_spread	Rules	B	Mechanical quarterly hedge, cost-capped at 1%/yr. Put-skew premium on solid empirical footing (Cole 2013).	~4
7	dealer_gamma_regime	Rules	B	GEX regime classification: long-straddle (negative GEX) vs IC (positive GEX). Sound thesis.	8–12
8	earnings_iv_crush	Rules	B	IV crush post-earnings is real and persistent (20–40% on avg); short iron condor 1-day hold captures the bulk.	40–60
9	earnings_post_drift	Rules	B	Post-earnings drift (SUE effect) is one of the most-replicated anomalies in academic finance.	40–60
10	ivr_credit_spread	Rules	B	Plain vanilla VRP harvest at $IVR \geq 50\%$. Boring is good — proven theta strategy.	12–18

Honorable mentions

- put_steal (AI, B) — NII regime signal + GBM survival on bull puts.
- vix_term_structure (AI, B) — VIX contango/backwardation regime classifier.
- rates_spy_rotation (Rules, B) — Simpler allocation version of #2.
- bull_put_spread (Rules, B-) — IVR-gated bull put. Solid but delta-approx instead of real BS.

The bottom (avoid until rewritten)

- **covered_call_ai** (AI, D) — Assignment modeling missing; backtest unreliable.
- **vol_calendar_spread**, **vol_term_structure_regime**, **oi_imbalance_put_fade**, **gamma_flip_breakout** (AI, D) — Incomplete or label-feature tautologies.
- **broken_wing_butterfly**, **calendar_spread**, **earnings_straddle**, **vol_arbitrage**, **gex_positioning**, **calendar_spread_vix** — heavy heuristic approximation.

The deep-dive below covers only the three originally-audited AI strategies. The other 35 were graded via triage in the table above.

TL;DR — Which AI strategy makes money?

hmm_regime is the only strategy with a credible path to real edge.

The other two have known flaws large enough to swamp any ML contribution.

The HMM strategy works because it is *not actually trying to predict the future*. It classifies the **current** vol regime (a well-documented stylized fact since Hamilton 1989, Ang–Bekaert 2002) and matches a defined-risk option structure to it. That is how real vol traders think. The other two strategies use ML to do something ML is bad at (predict direction or range) and then layer cost-modeling errors on top.

Ranking Table

#	Strategy	Thesis	Code Quality	ML Value	Risk & Live	Verdict
1	hmm_regime	Regime-switching is academically supported (Hamilton, Ang–Bekaert). Classifies <i>current</i> regime, doesn't predict.	Cleanest walk-forward; state-relabel handles label switching; slippage modeled; warm-start; forward-posterior gating. Minor: no margin tracking.	Genuine. HMM provides info a rules-based IVR/VIX gate doesn't.	All defined-risk. max_concurrent=1. Live wired with model + heuristic fallback.	DEPLOY
2	iron_condor_ai	"Predict range-bound next 45 days" is auto-correlated with IVR/VIX features the model already sees.	Already had one leakage fix. Still: no slippage, VIX-as-IV for any ticker, adaptive-delta logic runs backwards from Kelly.	Self-acknowledged ~0.5–0.9 Sharpe vs. rules IC baseline ~0.6–0.8. Marginal lift.	Defined-risk. max_concurrent=4. Margin tracked. Live IVR proxy disagrees with backtest.	PAPER ONLY
3	covered_call_ai	Thesis claims (delta × DTE) optimization; code is binary write-yes/no with two delta buckets. Label trivial.	No assignment modeling. Fake days_since_earnings. No commissions. ML model never used in live signal.	Model predicts "did stock not crash" — a directional/momentum signal in option-selling clothes.	Stock + short call. Loss path partly hidden by missing assignment. Live signal ignores trained model.	DO NOT USE

Trading Frequency

Strategy	Concurrent	Avg Hold	Entry Gate	Trades/Year
hmm_regime	1	12–25 days	$P(\text{state}) \geq 0.60$ & $\text{VIX} \leq 40$ & spot/forward agree	~12–18
iron_condor_ai	4	21–24 days	$P(\text{range}) \geq 0.50$ & $\text{IVR} \geq 0.20$ & $\text{VIX} \leq 38$	~30–50
covered_call_ai	1	10–18 days	$\text{IVR} \geq 0.30$ & $\text{days_since_earn} \geq 10$	~15–25

Important caveat: These are decision-frequency estimates, not fill-rates. On real chains expect 70–85% of these to actually fill at acceptable prices — bid/ask widens on illiquid strikes and IV mismatches against BS-priced theoretical credit/debit will reject some entries.

What “makes money” actually depends on

A strategy makes money in production when **all four** of these hold:

1. **Edge exists in the population** — there is a real, persistent inefficiency. (HMM ✓, IC_AI ~, CC_AI ×)
2. **The backtest faithfully simulates fills** — no theoretical mid-mark fantasies. (HMM ✓, IC_AI ×, CC_AI ×)
3. **The labels measure what the thesis says** — the ML is learning the right thing. (HMM N/A unsupervised, IC_AI ~, CC_AI ×)
4. **Live = backtest** — same features, same model path. (HMM ✓, IC_AI ×, CC_AI ×)

hmm_regime is the only one that passes all four.

Cross-cutting issues across all three

These hurt every “AI” strategy in the codebase regardless of rank:

- **VIX-as-IV is a SPY-only assumption.** Treating the VIX as the implied vol for *any* ticker’s options is fine for SPY/SPX-correlated names, materially wrong for single stocks. The description field claims ticker-genericity; the implementation doesn’t deliver it.
- **No bid/ask cost component.** Slippage parameters are inconsistent: HMM has `slip=0.05/leg`, IC_AI has 0, CC_AI has 0. A 4-leg IC closed at 50% of credit can be entirely consumed by 3–5¢/leg adverse fills.
- **Final saved model = last walk-forward fit.** All three save the most recent retrain’s model. That is typically trained on the most recent 9–12 months only — which may be the most regime-anomalous window. Consider refitting on full history before persisting for live use.
- **Tree-based GBM with StandardScaler.** Both `iron_condor_ai` and `covered_call_ai` pipeline a scaler ahead of GBM. Trees are scale-invariant. Harmless but cargo-cult.

Recommended next moves

In order of leverage:

1. **Promote `hmm_regime` to first-class.** Add reserved-margin tracking, thread per-ticker *q*, parameterize the bull-put / long-put `dte_ex` exits. Paper-trade for 60–90 days on SPY before risking capital.
2. **Park `iron_condor_ai` until** you have added bid/ask slippage, threaded per-ticker IV (read from `option_snapshots`), and either fixed or A/B tested the inverted adaptive-delta logic.
3. **Rewrite `covered_call_ai` before quoting any number from it.** The assignment-modeling gap is a correctness bug, not a refinement. Until then its backtest equity curve is fiction.
4. **Treat the “rules” IC as the honest baseline.** If `iron_condor_ai`’s post-fix Sharpe is 0.5–0.9 and the rules IC already does ~0.6–0.8 (per the IC_AI audit comments), the ML layer is not earning its complexity budget. That is a defensible reason to *remove* the AI variant, not improve it.

Caveats on this ranking

- Scores reflect what I can verify by reading source code on **2026-05-19**, not live performance.

- I have not run any of the backtests; ratings are about *whether the backtest is trustworthy*, not what its equity curve shows.
- The “Expected Real Sharpe” column in the source markdown is a wide range, not a forecast. It reflects how much I would haircut the headline backtest after correcting the issues listed.
- An “edge candidate” still needs walk-forward out-of-sample validation, paper trading, and small-size live validation before it earns real capital.

Note: this ranking covers only the three *_ai / hmm_* strategies originally audited.
A broader pass over all 18 StrategyType.AI_DRIVEN strategies is in progress separately.